J.M.S

COPYRIGHT RESERVED

BCA (Sem-III) – Data Structure (301) Core-V

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all group as directed.

Group—A

(Objective Type Questions)

1. Choose the correct answer : $2 \times 10 = 20$

(i) In a stack, the insertion operation is called:

A) Enqueue B) Push

C) Pop D) Delete

(ii) The last node of a circular linked list points to:

A) NULL B) Previous node

C) First node D) Random node

(iii) Which searching technique requires the data to be sorted?

A) Linear Search B) Sequential Search

C) Binary Search D) Hash Search

(iv) Which data structure is used to evaluate postfix expressions?

A) Queue B) Stack

C) Linked List D) Tree

(v) In a Priority Queue, elements are removed according to:

A) FIFO principle B) LIFO principle

C) Their priority D) Random order

(vi) An AVL tree maintains balance by ensuring that:

A) Every node has two children

B) The balance factor is –1, 0, or +1

C) All leaves are at the same level

D) The tree is complete

(vii) Which expression notation places the operator after its operands?

A) Prefix B) Infix

C) Postfix D) Unary

(viii) Which of the following traversal methods visits the root node last?

A) Preorder B) Inorder

C) Postorder D) Level Order

(ix) A two-dimensional array is commonly used to represent:

A) Stack B) Queue

C) Matrix D) Tree

(x) The primary purpose of a hash function is to:

A) Sort records

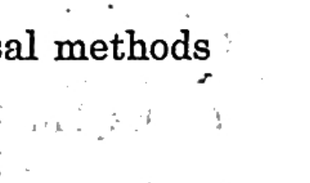
B) Compress files

C) Map keys to table indices

D) Balance trees

Group – B

(Short Answer Type Questions)



Answer any **four** of the following : 5×4=20

2. Define stack and explain its basic operations with examples.

3. Explain memory allocation in arrays.

4. Explain doubly linked lists and their advantages.

5. What is a circular queue? Explain its advantages.

6. Explain recursive solution of Fibonacci series.

7. Explain Binary Search Tree (BST) with examples.

8. Compare Linear Search and Binary Search.

Group – C

(Long Answer Type Questions)

Answer any **two** of the following : 15×2=30

9. Describe insertion, deletion, and traversal operations in singly linked lists. Write algorithms for each operation and analyze their time complexity.

10. Explain Linear Search and Binary Search algorithms with examples. Compare their efficiencies and discuss their time complexities.

11. Define tree and explain tree terminology. Discuss different types of trees and their applications in computer science.

12. Explain the concepts of single-dimensional and multi-dimensional arrays. Discuss their memory representation, advantages, disadvantages, and applications with suitable examples.

———— × ————